

Lat: 7.509N Lon: 99.617E Elev: 20 StdP: 101.08 Time Zone: 7.00 (E07) Period: 94-19 WMO: 485670 WBAN: 99999

TRANG, Thailand

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.8	22.6	18.6	13.5	29.9	19.3	14.1	29.1	8.0	29.8	7.2	30.0	0.1	30	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.6	35.6	25.7	34.8	25.7	34.1	25.8	27.6	32.0	27.3	31.6	27.1	31.4	3.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.2	29.6	26.2	21.7	29.3	26.0	21.4	29.2	88.1	32.3	86.7	31.9	85.5	31.6	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.3	6.3	5.4	DB	20.4	36.5	1.2	1.0	19.5	37.2	18.8	37.8	18.1	38.3	17.2	39.1	
			WB	19.4	28.8	1.1	0.6	18.6	29.2	18.0	29.5	17.3	29.9	16.5	30.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	27.4	28.1	28.9	29.4	28.9	28.4	28.0	28.0	27.9	27.7	27.4	27.1
	DBStd	1.28	1.05	0.97	1.01	1.13	1.15	1.10	1.07	1.11	1.09	1.03	1.24	1.16
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6602	539	507	585	581	586	551	559	559	536	548	522	530
	CDD18.3	3561	281	273	327	331	328	301	301	301	286	290	272	272
	CDH23.3	35731	2752	2958	3686	3654	3524	3131	3014	2979	2678	2530	2399	2424
CDH26.7	14020	1025	1302	1771	1704	1461	1219	1143	1106	943	822	757	766	
Wind	WSAvg	1.5	2.3	2.4	1.8	1.1	1.1	1.2	1.2	1.3	1.2	1.1	1.4	2.0
Precipitation	PrecAvg	2273	64	32	48	103	265	240	252	263	358	285	241	124
	PrecMax	2923	466	93	261	267	522	555	527	504	527	466	429	288
	PrecMin	1775	0	0	0	3	57	84	91	126	203	130	76	9
	PrecStd	334	123	33	60	64	108	113	104	99	100	100	89	79
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	35.1	36.4	37.1	36.0	34.2	33.9	33.9	33.8	33.8	33.4	33.1
		MCWB	25.2	25.0	25.1	25.9	26.8	26.5	26.3	26.2	26.3	26.1	26.0	25.4
	2%	DB	32.7	34.1	35.2	35.8	34.6	33.6	33.1	33.1	33.0	32.8	32.6	32.2
		MCWB	24.9	24.7	25.0	26.1	26.8	26.4	26.1	26.1	26.1	26.0	25.9	25.2
	5%	DB	32.0	33.3	34.6	34.8	33.6	33.0	32.5	32.5	32.2	31.9	31.9	31.6
		MCWB	24.5	24.5	24.9	26.2	26.8	26.4	26.1	26.1	26.0	26.0	25.8	25.1
	10%	DB	31.2	32.6	33.8	33.8	32.6	32.1	31.8	31.7	31.2	30.9	30.8	30.7
		MCWB	24.4	24.4	25.0	26.2	26.7	26.4	26.2	26.1	26.1	26.0	25.7	24.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.3	26.5	27.1	27.9	28.1	27.9	27.6	27.4	27.4	27.2	27.1	26.6
		MCDB	30.9	32.2	32.6	33.3	32.6	31.7	31.4	31.3	31.4	31.1	30.9	30.6
	2%	WB	25.8	25.9	26.6	27.4	27.6	27.5	27.1	27.0	26.9	26.7	26.6	26.1
		MCDB	30.6	31.6	32.0	32.6	32.1	31.4	31.0	30.9	30.8	30.6	30.5	30.2
	5%	WB	25.4	25.4	26.2	27.0	27.2	27.1	26.8	26.7	26.6	26.4	26.2	25.7
		MCDB	30.0	31.0	31.6	32.2	31.7	31.1	30.6	30.5	30.4	30.2	30.1	29.6
	10%	WB	25.0	25.0	25.8	26.6	26.9	26.7	26.5	26.4	26.2	26.1	25.9	25.3
		MCDB	29.4	30.4	31.0	31.6	31.3	30.6	30.2	30.1	29.9	29.8	29.7	28.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.6	10.2	9.6	8.1	7.5	7.5	7.3	7.2	7.1	6.7	6.8
		MCDBR	9.2	10.8	11.6	10.9	9.4	8.8	8.6	8.6	8.5	8.4	8.4	8.7
		MCWBR	2.9	3.1	3.1	2.9	3.0	3.0	2.9	2.8	2.8	2.8	2.8	2.9
	5% WB	MCDBR	8.1	9.8	10.2	9.7	8.3	7.9	7.8	7.7	7.7	7.6	7.4	7.2
MCWBR		2.8	3.2	3.2	3.2	3.0	3.1	3.1	2.9	2.9	2.9	2.8	2.8	
Clear-Sky Solar Irradiance	taub		0.469	0.478	0.487	0.473	0.452	0.457	0.460	0.455	0.445	0.447	0.457	0.465
	taud		2.258	2.209	2.219	2.263	2.353	2.368	2.345	2.345	2.370	2.364	2.326	2.287
	Ebn at Noon		847	851	844	843	843	828	829	847	868	868	852	842
	Edh at Noon		139	149	149	141	125	122	125	128	126	126	129	133
All-Sky Solar Radiation	RadAvg		4.96	5.88	5.93	5.73	5.08	4.88	4.82	4.94	4.78	4.54	4.18	4.15
	RadStd		0.40	0.36	0.56	0.32	0.29	0.29	0.24	0.19	0.16	0.33	0.30	0.42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	+0.19	+0.25	N/A	N/A	+0.19	+0.21	N/A	N/A	+74	+75